



1. Introduction

The EP560 SD slave controller from Eureka Technology can be configured to support SD 2.0 standard and/or MultiMedia Card (MMC) 4.2 standard. MMC is very similar to SD/SDIO standard so Eureka's EP560 SD slave controller can be delivered in any of the following 3 configurations:

1. SD/SDIO only support.
2. MMC only support.
3. SD/SDIO and MMC support, selected by hardware mode pin.

This document describes the additional features of MMC support for the EP560 IP core. For the basic features of that IP core that is common to both standards, please refer to the EP560 data sheet.

2. Signal Descriptions

The major differences between SD2.0 and MMC4.2 is that MMC supports 8-bit data transfer and stream mode data transfer (CMD11 and CMD20). The user interface of the EP560 is versatile enough that it can support both standards without change. The majority of the I/O pins of the EP560 remains unchanged with MMC. The following table lists the few signals that are affected by MMC4.2.

Table 1: Signals for MMC Support

Name	Type	Descriptions
DT[7:0]	I/O	Data line. When MMC support is added, the data line changes from 4 to 8 bit to support 8 bit data transfer.
ECSD_A[7:0] - ECSD_Q[7:0]]	I	Extended CSD read only data, static input. Extended CSD is a register specific to MMC mode. The register contains mostly read only data and the read only values are provided to the core through this static input. User can hardware these values according to the characteristics of the card.

**Table 1: Signals for MMC Support**

Name	Type	Descriptions
RD_THD[2:0]	I	Read Threshold for stream data read, static input. In stream read mode, in order to prevent data under-flow, the user may want to buffer enough data inside the EP560 before the first data is returned to the SD card interface. The encoding of the RD_THD is the following: 000: Buffer 16 words before sending data to host. 001: Buffer 32 words before sending data to host. 010: Buffer 64 words before sending data to host. 011: Buffer 128 words before sending data to host. 111: No Buffer, send data to host as soon as first word arrive from user interface. Others: Reserved
SEL_MMC	I	Select MMC, static input. This signal is present if the core supports both SD mode and MMC mode. This signal is not needed if the core supports only SD or MMC but not both. This signal is “1” to indicate MMC mode and “0” to indicate SD mode. This signal must remain unchanged after reset.

If the EP560 is setup for MMC only with no SD support, the following static inputs related to SD only registers are removed.

Table 2: Signals Not Needed for MMC Only Support

Name	Type	Descriptions
CCCR[144:0]	I	Static input. Lower 145 bits of Card Common Control Registers read only values for SDIO
CCCR_OUT [145:17]	O	Output from CCCR register, reflect the values of the corresponding bits in the CCCR register. Only the following bits which are writable by the host have valid information. All other bits are hardwired to “0”. The valid bits are: 145, 143:128, 107:104, 97, 69, 63, 61, 57:56, 51:48, 39:32, 23:17.

**Table 2: Signals Not Needed for MMC Only Support**

Name	Type	Descriptions
CCCR_UPP ER[127:0]	I	Static input. Upper most 128 bits of Card Common Control Registers read only values for SDIO
FBR1[143:0]	I	Static input. Function Basic Register read only values for SDIO Function 1.
OCR_SDIO[23:0]	I	Static input. Operation Condition Register read only values, for SDIO.



3. Functional Descriptions

3.1 8-bit Data Transfer

The biggest advantage of using MMC is 8-bit data transfer which doubles the data transfer of SD card. When the MMC feature of the EP560 is used, the core automatically supports 8-bit data transfer. The core has 8 bits on the card's DT line instead of the usual 4. Under software control by the host, the card can be switch between 1, 4 and 8-bit data transfer.

To the user interface and user logic, 8-bit data transfer is completely transparent. Since all the data from the card interface is serially shifted in to the card and transferred to the user logic through a 32-bit data bus, the user interface protocol remains unchanged regardless of the data width on the card side interface. The only observable differences by the user logic is that ADS# can be asserted earlier between blocks of data because each block takes less time to process.

3.2 Bus Testing Procedure

The MMC bus uses the CMD14 and CMD19 commands to detect the data bus width of the device. The test procedure is initiated by the MMC host and the slave controller automatically respond to the MMC host to signal 8-bit data line connection. The test procedure is built-in to the MMC slave controller and no user interaction is required. The MMC slave controller ensures that the highest data bandwidth is maintained on the MMC bus with the widest data path allowed by the system.

3.3 Stream Mode Data Transfer

Stream mode data transfer is a unique feature of the MMC bus. In stream data mode transfers data continuously across block boundary. The MMC host issues CMD11, CMD12 and CMD20 commands for stream read and stream write data transfer. The MMC slave controller handles stream data mode automatically. In stream mode operation, data is still read or written from the user logic through the normal ADS#/RDY# bus protocol. In other words, stream mode is transparent to the user logic.

Because data bandwidth on the MMC bus stream mode and user logic may be different, the MMC slave controller uses the internal data buffer as a FIFO to maintain steady data flow.

On write operation, the MMC slave stores the write data from the host in the data buffer and then write to the user logic. It starts to transfer write data to the user logic when 4 words of data has been accumulated in the data buffer. It uses burst write of 4 words. If the data transfer rate on the user interface is slow, data starts to accumulate on the data buffer. If 16 or more words has accumulated in the data buffer, the MMC slave controller uses burst of 16 words to write data out to the



user logic.

3.4 Stream Read Threshold

On read operation, the MMC slave controller issues burst read access of 16 words to read data from the user logic and deposit into its data buffer. Data is then returned to the MMC host at the steady rate of the stream read mode. Buffer overflow is prevented by the MMC slave controller. It does not issue burst read to the user logic if the data buffer is about half full.

Data buffer under-flow may occur if the user logic read latency is very long. To avoid data underflow, it is desirable to accumulate several words of data in the data buffer. Streaming data back to the host starts only after the number of data words in the buffer reaches the threshold. The user logic can control the buffer threshold by using the RD_THD static input. In systems with even read latency, a small threshold is preferred. In systems with large read latency when crossing memory physical boundary, a large threshold is preferred.

3.5 MMC Specific Register

The Extended CSD is an MMC specific register. This register is internal to the MMC slave controller. The register contain mostly read only data fields that describes the characteristic of the MMC card. User logic specifies these read only values through the following static inputs. These inputs can be hardwired to the desirable values by the user or connected to user registers.

Table 3: Extended CSD Static Inputs

Static Input	Extended CSD Field
ECSD_A[7:0]	S_CMD_SET
ECSD_B[31:0]	SEC_COUNT
ECSD_C[7:0]	MIN_PERF_W_8_52
ECSD_D[7:0]	MIN_PERF_R_8_52
ECSD_E[7:0]	MIN_PERF_W_8_26_4_52
ECSD_F[7:0]	MIN_PERF_R_8_26_4_52
ECSD_G[7:0]	MIN_PERF_W_4_26
ECSD_H[7:0]	MIN_PERF_R_4_26
ECSD_I[7:0]	PWR_CL_26_360
ECSD_J[7:0]	PWR_CL_52_360

**Table 3: Extended CSD Static Inputs**

Static Input	Extended CSD Field
ECSD_K[7:0]	PWR_CL_26_195
ECSD_L[7:0]	PWR_CL_52_195
ECSD_M[7:0]	CARD_TYPE
ECSD_N[7:0]	CSD_STRUCTURE
ECSD_O[7:0]	EXT_CSD_REV
ECSD_P[7:0]	CMD_SET_REV
ECSD_Q[7:0]	ERASE_MEM_CONT

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